

EDUCATION

Heritage High School, Frisco, TX

Class of 2027

GPA: 4.00 UW / 5.361 W · Rank 11/481 · ACT 36 · AP Scholar with Distinction

AP exams: 9 taken, a 5 on all 9 — Computer Science A, Statistics, Physics 1, Seminar, English Language & Composition, U.S. History, World History, Psychology, Human Geography

Coursework: AP Calculus BC, AP Physics 2, AP Physics 1, AP Computer Science A, AP English Language, AP English Literature, AP Government, AP Economics, Computer Science III, Mobile App Programming, Spanish 2

Competitions: Semifinalist, Diamond Challenge 2026 (X-Step) · Finalist, Texas High School Ideas Challenge at Texas A&M (X-Step, May 2026)

mcferrin.tamu.edu/program/texas-high-school-ideas-challenge · 1st, TechLit Bridging the Gap Hackathon (ShopSmart, Aug. 2025) · Innovation Prize, PeddieHacks (235 participants) · 3rd, Hack4Humanity

RESEARCH

Predicting Cognitive Function Using Transformer-Derived Speech Representations and Longitudinal Coherence Features — *First Author*

May 2025–Present

Mentor: Joe Xiao (PhD student) · Under peer review, *Frontiers in Aging Neuroscience* — submitted August 04, 2026 · Preprintmedrxiv.org/content/10.64898/2026.08.04.26359730v1

- Question:** can a patient's speech across clinic visits forecast their next Mini-Mental State Exam (MMSE) score? Framed as supervised regression on the DementiaBank Pitt Corpus (173 longitudinal samples, 126 patients) using only earlier visits.
- Representation:** to capture how discourse organization changes with decline, extracted 12 per-visit linguistic features (POS fragment ratio, named-entity rate, 50-token MATTR, filler rate, perseveration) and Sentence-BERT coherence features (adjacent-sentence cosine similarity, global coherence, semantic slope, topic drift, embedding variance), plus visit-to-visit Δ features.
- Model:** reduced 384-d SBERT embeddings to 20 PCA components, added 9 clinical/demographic covariates (53 dimensions), and tuned LightGBM with Optuna/TPE over 200 trials (RobustScaler, correlation pruning, permutation-importance selection).
- Evaluation:** found patient-level leakage (one patient on both sides of a split) and rebuilt evaluation with patient-grouped 5-fold GroupKFold, augmented copies kept in their patient's fold, original samples only scored: R^2 0.840 $\pm$ 0.015, RMSE 2.75, MAE 2.02 MMSE points. Ablation: clinical history carries most of the signal (clinical-only R^2 0.72, SBERT-only 0.46, linguistic-only 0.12); speech adds a smaller gain (0.76 combined); non-nested tuning is disclosed as an optimism limit. Code github.com/ddevatha1/dementia-mmse-prediction

X-Step: Four-Site Plantar-Pressure Insole — *ML Lead, 2nd of 5 Authors*

Aug. 2025–Present

UC Santa Cruz AIEA Lab (Prof. Leilani Gilpin) aiea-lab.github.io · MIT CSAI co-authors · Under review, *IEEE-EMBS IECBES*2026 · deeptanshudevatha.com/research/x-step

- Question:** dense in-shoe pressure arrays are costly and lab-bound; what can a cheap four-sensor insole measure? The paper evaluates X-Step as a measurement instrument, not a clinical classifier. I implemented the 59-feature representation (per-site peak/mean/CV, pressure-time integral, cadence, stance ratio, anterior load share) from four FSR402 sensors at 25 Hz over BLE from an ESP32, with a 540-parameter logistic-regression head.
- Designed a sparse-vs-dense study (149 recordings, 15 adults, public 32-cell insole): four sites keep waveform shape (median r = 0.72–0.99) but not magnitude (peak-to-peak r = 0.43–0.74); center-of-pressure tracking keeps anteroposterior motion (r = 0.81) but loses mediolateral (r = 0.01).
- Quantified subject-identity leakage on a synthetic 24-subject cohort (macro-F1 0.885 subject-grouped vs 0.931 ungrouped) and documented the generalization limit.

PROJECTS & LEADERSHIP

ShopAI — *Creator & Sole Developer*

2025–Present

- Prices differ by store and no retailer offers an API built for comparison, so ShopAI searches nine chains at once and plans the trip. Built end to end: React Native/Expo client and an Express/TypeScript backend on AWS EC2 behind Nginx, with adapters that turn unlike sources (OAuth2 REST, cookie-session GraphQL, Playwright, server-rendered HTML) into one product schema. Live on the App Store. apps.apple.com/app/shopai/id6797083809
- Concurrent fan-out search under a shared response budget with per-store failure states, query-correction/relevance/ranking, and a per-ZIP auth warm-up state machine. A deterministic planner enumerates all 127 non-empty subsets of the 7 live-data stores (exhaustive search is cheap, so no heuristic) for cheapest, fastest, fewest-stop and weighted plans, with zero LLM calls. deeptanshudevatha.com/projects/shopai

KidCode — *Founder & Lead Instructor* · Frisco, TX

2024–Present

- Founded a 3-year Python workshop serving 124 students across 10 Frisco neighborhoods; recruited peers to help teach. Wrote the 120-class-day, 20-project curriculum: control flow, file I/O, JSON, exceptions, OOP/inheritance and assert-based tests, then spaCy NLP (tokenization, lemmatization, POS tagging, sentence segmentation, NER, Matcher patterns, a rule-based chatbot). deeptanshudevatha.com/projects/kidcode
- To test learning, wrote 13 pre/post assessment pairs (same skills, different code): across 104 students the mean score rose 41.7% $\rightarrow$ 58.8% (+17.1 pp; 99 improved, 4 unchanged, 1 declined; no comparison group). Students completed 1,214 working programs, 572 using spaCy. deeptanshudevatha.com/projects/kidcode/assessment

X-Step — *Co-Founder, ML & Mobile Lead*

Aug. 2025–Present

- X-Step is a student startup building a smart insole for people at risk of diabetic foot ulcers, which form where pressure builds up unnoticed. The team's ESP32-C3 insole streams four pressure sensors over BLE; I wrote the software: the React Native/Expo app (BLE client, live pressure heatmap, daily risk score, alerts, trends, guided routine) and the earlier Flask/SQLite + LLM health-assistant backend. deeptanshudevatha.com/projects/x-step
- Trained two CNN prototypes on public data: EfficientNet-B2 on 9 foot-pressure heatmap postures (98.3%, random 80/20 split) and ResNet-50 on four wound-severity grades. Intended flow: an ulcer photo's severity score plus care habits go to an LLM that drafts routines. Bench-stage prototype; no wearer data, no clinical validation.

Ayusha Nilayam — *Programming & Math Instructor* · Hyderabad, India (remote)

Nov. 2025–Jul. 2026

- Taught 38 weekly Zoom sessions of Python and math to 11 children at a Hyderabad children's home, adapting to one shared laptop, unreliable internet and a Telugu–English language gap; then two weeks in person. deeptanshudevatha.com/projects/ayusha-nilayam

UT Dallas NLP with Python Workshop — *Participant*

Summer 2025

Dr. Chris Davis · Built a rule-based text simplifier: dependency parsing, lexical substitution, clause splitting, passive-to-active rewriting.

TECHNICAL SKILLS

ML/NLP: LightGBM, scikit-learn, Optuna, Sentence-BERT, spaCy, PyTorch, PCA, GroupKFold, permutation importance · Languages: Python, TypeScript/JavaScript, Java, SQL

Systems: React Native/Expo, Node.js/Express, Flask, REST, GraphQL, OAuth2, Playwright, AWS EC2, Nginx · Embedded: ESP32, BLE, FSR sensing